

POKHARA UNIVERSITY
GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

LABORATORY REPORT
AGILE SOFTWARE DEVELOPMENT LAB

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NAME OF EXPERIMENT

Exploring Agile project management tools (Jira, Asana, Trello, ClickUp, and Zoho Projects).

AIM

To study and use the basic features of Agile project management tools for planning, organising, assigning, tracking, and managing software development projects.

THEORY

Agile project management is an iterative and incremental way of running software projects. In contrast with the traditional waterfall model, Agile stresses flexibility, ongoing collaboration, customer feedback, and delivery of working software in short, frequent increments called sprints. Agile teams are self-organising and cross-functional and rely on openness and communication.

To put frameworks such as Scrum and Kanban into practice, teams use dedicated project management tools that organise work, track progress, and support collaboration on a single shared platform. The tools explored in this experiment are described below.

1. Jira (Atlassian): Jira is among the most widely used Agile tools in the industry. Built for issue tracking and Agile management, it supports both Scrum and Kanban and lets teams write user stories, groom backlogs, plan sprints, assign tasks, and produce reports such as burndown and velocity charts. Its strong customisation and integrations suit teams of any size.
2. Asana: Asana is a cloud-based tool centred on team collaboration and task organisation. Users create projects, break them into tasks and subtasks, assign owners, set due dates, and attach files. It offers list, board, calendar, and timeline views, and its clear interface and notifications keep everyone aligned.
3. Trello: Trello is a lightweight, visual tool based on Kanban. Tasks are cards that move across lists such as To Do, In Progress, and Done. It is very easy to use and well suited to small teams or personal tracking, with support for checklists, labels, due dates, attachments, and power-up integrations.
4. ClickUp: ClickUp is an all-in-one productivity and project management platform that aims to replace several tools at once. It provides task management, sprint planning, goal setting, time tracking, document collaboration, automation, and customisable dashboards, and it supports both Scrum and Kanban.
5. Zoho Projects: Zoho Projects is an online tool for planning projects, assigning work, tracking issues, managing documents, and monitoring timelines. It offers Gantt charts, task boards, timesheets, and discussion forums, and integrates well with the wider Zoho suite.

Collectively these tools serve the core needs of Agile teams: transparency so everyone sees the work, accountability through clear assignment, adaptability so priorities can shift, and continuous improvement through reporting and retrospectives.

OBJECTIVES

- To understand the core principles of Agile project management.
- To explore and compare Jira, Asana, Trello, ClickUp, and Zoho Projects.
- To create and manage projects, tasks, and user stories in these tools.
- To assign work to team members and set priorities and deadlines.
- To track progress using Kanban boards, Scrum boards, and dashboards.
- To understand sprint planning, backlog management, and progress monitoring.

ALGORITHM

1. Choose one of the Agile tools (Jira, Asana, Trello, ClickUp, or Zoho Projects).
2. Create an account or sign in to the chosen platform.
3. Create a new project with a name, description, and methodology (Scrum or Kanban).
4. Populate the backlog with tasks, user stories, or work items.
5. Group tasks logically using epics, milestones, or categories.
6. Assign each task to a team member and set a priority (High, Medium, Low).
7. Set due dates and estimated effort for each task.
8. Move tasks into board columns such as To Do, In Progress, Review, and Done.
9. For Scrum, create a sprint and pull selected backlog items into it.
10. Update task statuses as work progresses.
11. Monitor project health with dashboards, reports, and burndown charts.
12. Run reviews or retrospectives using the collaboration features.
13. Record observations and compare the tools.

IMPLEMENTATION

The following activities were carried out (Trello and Jira were used hands-on, while the others were studied conceptually).

1. Project creation: a project named "Agile Task Manager" was created on the platform.
2. Backlog creation: several user stories and tasks were added, each with a title, description, and acceptance criteria.
3. Task assignment: tasks were assigned to team members according to role and expertise.
4. Prioritisation: each task was given a priority so that critical work came first.
5. Sprint planning: a two-week sprint was planned and selected backlog items were moved into it.
6. Board management: cards were advanced from To Do to In Progress, then Review, then Done.
7. Progress tracking: the dashboard was used to watch completion rates and workload and to spot bottlenecks.
8. Collaboration: team members commented, attached files, and updated statuses in real time.

OUTPUT

The Agile project management tool was used successfully to run the project activities.

- The project was created with all required configuration.
- Tasks and user stories were added to the backlog effectively.
- Team members were assigned tasks with appropriate priorities.
- Task progress was tracked transparently on the Kanban/Scrum board.
- Project status was monitored through real-time dashboards and reports.
- Collaboration and communication among the team improved noticeably.
- Bottlenecks were identified and work was reprioritised efficiently.

RESULT

The Agile project management tools were used successfully. They helped organise tasks, improve collaboration, monitor progress, and manage development work in a structured way. Hands-on practice with Trello and Jira, together with the conceptual study of Asana, ClickUp, and Zoho Projects, gave a well-rounded view of the Agile tooling landscape.

CONCLUSION

This experiment provided both practical and theoretical knowledge of Agile project management tools including Jira, Asana, Trello, ClickUp, and Zoho Projects. These tools are indispensable for modern Agile teams: they simplify planning, strengthen task management, encourage collaboration, and enable real-time progress tracking. By choosing the tool that best fits their needs, teams can markedly improve productivity, transparency, and their ability to deliver quality software on time. Familiarity with them is essential for any professional working in an Agile environment.